# AI Crypto Hedge Fund

A Risk-First, Reproducible Research MVP

Causal Data → Signal Agents → Risk & Portfolio → Next-Open Execution → Evidence

| | | |
|---|---|---|
| 5 | 120 | 0 |
| Research Levels | Pairs Scored | Future-Label Violations |

**Educational historical research MVP.** This project proves a disciplined research and control architecture; it does not claim persistent alpha or live-trading readiness. Long-only, unlevered, daily spot, USDT cash.

Source: frozen committed artifacts; current accepted final-test lock c33b5eb...

# One Architecture, Increasing Portfolio Complexity

## UNIVERSE EXPANSION

Levels 1-2 — **BTC / USDT**

↓

Levels 3-4 — **7 Liquid Assets**

↓

Level 5 — **120 Eligible / Scored -> Top 25**

## INVESTMENT MANDATE

- Spot market only
- Long-only positions
- No leverage
- USDT cash explicit
- Daily rebalancing horizon
- Next-open execution

**Why this progression?**

BTC/USDT gives a controlled single-asset comparison. 5-7 assets make covariance estimation interpretable. 120+ pairs prove scalability without holding every pair. One USDT quote simplifies accounting.

⚠ Appropriate for a research MVP; not a point-in-time institutional universe because delisted-market reconstruction and order-book history are absent.

Source: frozen committed artifacts; current accepted final-test lock c33b5eb...

# Diverse Models Propose; Deterministic Controls Decide

**SMA**
Trend, rule-based

**AR / GARCH**
Return + volatility

**Logistic**
Linear ML

**HGB**
Nonlinear ML

**Equal-Weight Aggregator**

**Risk / Portfolio / Execution**

## CAUSAL CLOCK

Bar *t* completes → Features / decision → Execute open($t$+1) → Evaluate open($t$+2)

## KEY PARAMETERS

- 20 causal features
- Target > 0.20% return
- Monthly ML refit
- Daily econ refit
- 0 future-label flags

## MODEL RATIONALE

| SMA | Transparent rule-based baseline; easy to interpret and debug |
|---|---|
| AR / GARCH | Classical expected-return and volatility econometrics |
| Logistic | Interpretable, low-variance ML baseline for small daily samples |
| HGB | Captures nonlinear interactions with controlled complexity |

Source: frozen committed artifacts; current accepted final-test lock c33b5eb...

# Risk Is the Control Plane

Models cannot place orders; risk can cap, reject, or move to cash

**Agent Proposals**

↓

**PRE-RISK**
freshness validity
liquidity model health

↓

**Portfolio Proposal**

↓

**POST-RISK**
weights concentration
turnover cash reconcile

↓

**Approve / Cap / Cash / Kill**

## METRIC GROUPS MONITORED

### Market / Tail
Volatility, VaR, CVaR, maximum drawdown

### Trading
Turnover, fees, slippage, liquidity, capacity

### Portfolio
Exposure, concentration, effective N

### Model / Data / System
Drift, stale inputs, disagreement, incidents, runtime

**Deterministic gates control portfolio state.** GARCH and realized volatility inform risk estimates, but pre/post-risk gates - not AI free-form reasoning - make the final decisions.

**Note:** "System health = OK" means the pipeline operated correctly; it does not mean the strategy was profitable.

Source: frozen committed artifacts; current accepted final-test lock c33b5eb...

# Portfolio Optimality Is Objective-Dependent

## CANDIDATE METHODS

### Equal Weight
Robust benchmark; no estimation error

### Inverse Volatility
Reduces allocation to volatile assets

### Minimum Variance
Uses covariance; suffers estimation error

### CVaR / Downside
SELECTED

Focuses on tail losses; selected on validation

**Selection:** Level 3 used the prior 12 months only. CVaR was selected on 2024 validation by the frozen risk-adjusted criterion - not because it was best in 2025.

## TRADING PATH

### Seven Assets
BTC, ETH, SOL, BNB, XRP, DOGE, PEPE

### Estimation Window
Trailing 12 months (2024-01 to 2024-12)

### Execution Flow
Target weights -> risk approval -> next-open orders -> ledger

Source: frozen committed artifacts; current accepted final-test lock c33b5eb...

# Adaptation Must Earn Its Turnover

## REBALANCE APPROACHES

### Calendar Monthly

FROZEN

Time-based; selected under validation MDD <= 25% and turnover <= 6.0

### Drift Threshold

5% weight deviation trigger; not selected post-hoc

### Signal / Risk Trigger

15% score trigger + risk overlay; not selected post-hoc

**Rule:** More active policies may look better in the exposed final year, but cannot be selected post hoc. The frozen selection stands.

## LEVEL 5: LARGE UNIVERSE FUNNEL

163 stored symbols

↓

120 eligible / scored

↓

Top 25 selected

↓

Max 5% per asset

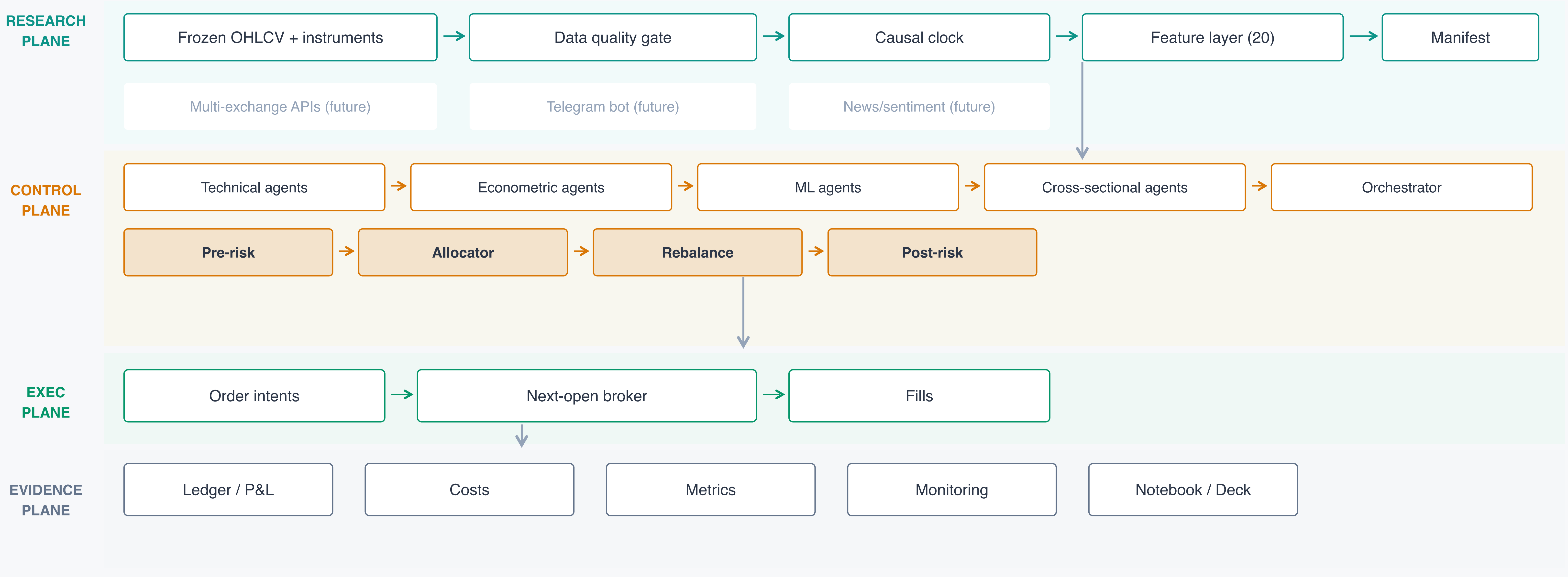
↓

Inverse-vol allocation

**Deterministic factor ranker:** momentum, volatility, drawdown, liquidity. Not a fitted ML model. A transparent scalable ranker replaces an unstable 120-asset covariance optimizer.

Source: frozen committed artifacts; current accepted final-test lock c33b5eb...

# A Shared Panel-Native Engine




*Grey dashed modules = future adapters (not active): multi-exchange APIs, Telegram controls, news/sentiment ingestion.*

Source: frozen committed artifacts; current accepted final-test lock c33b5eb...

# The Final Year Was an Exam, Not a Tuning Set

## TIMELINE

| 2021-2023 | 2024 | PRETEST LOCK | 2025 |
|---|---|---|---|
| TRAIN | VALIDATION & SELECTION | c33b5eb... | FROZEN FINAL TEST |

## DATA QUALITY

| 158,511 | 163 | 0 | 0 | 0 | Python 3.11 + uv |
|---|---|---|---|---|---|
| Asset-day rows | Stored symbols | Duplicate symbol-days | Null OHLCV fields | Future-label violations | Reproducible environment |

## REPRODUCIBILITY COMMANDS

```
$ uv sync --frozen
$ make validate-data && make lint && make test
$ make notebook-full && make presentation && make release-verify
```

**Causal clock:** completed bar $t$ -> features / decision -> execute open($t$+1) -> evaluate open($t$+2) | **Notebook loads frozen artifacts; does not rerun final test**

Source: frozen committed artifacts; current accepted final-test lock c33b5eb...

# The Architecture Generalized; The Alpha Did Not

## KEY FINDINGS

### Level 2: Ensemble

**-0.6%** vs BTC bench -5.4%

Defensive relative behavior; no persistent alpha

### Level 5: Large Universe

**-28.0%** net | bench -45.2%

~$110.9k costs | 4,924% turnover | gross -18.6%

### Robustness

95% CI includes poor outcomes; circular-shift p > 0.05. Validation signals did not generalize.

**Research conclusion:** Negative results strengthen credibility because they were not hidden or retuned away. The strongest evidence is reproducibility, leakage control, architecture, and scalability - not proven profitability.

*All results net of fees and slippage. Validation and final evidence are shown separately. Post-hoc final diagnostics are marked as such and did not drive selection.*

Source: frozen committed artifacts; current accepted final-test lock c33b5eb...

# What We Proved, What We Did Not, and What Comes Next

## PROVED

- ✓ Reproducible shared engine
- ✓ Causal next-open protocol
- ✓ Real ML / econometric fits
- ✓ Typed agent interaction
- ✓ Deterministic risk control
- ✓ 100+ pair scalability

## NOT PROVED

- ✗ Persistent alpha
- ✗ Live execution readiness
- ✗ Institutional capacity
- ✗ Point-in-time delisted universe
- ✗ Robust Level 5 economics

## NEXT VERSION

*New lock required*

- ▶ Calibrated probabilities
- ▶ Validation-selected ensemble weights
- ▶ Pooled Level 5 ML ranker
- ▶ Broader validation window
- ▶ Spread / order-book impact
- ▶ Multi-CEX + reconciliation
- ▶ Telegram / news adapters

**The value of this MVP is disciplined research and control - not a profitability claim.**

Lock: c33b5eb396f6...43af5e | Base tag: stage/09-level-5-100pairs | Public URL: owner to verify